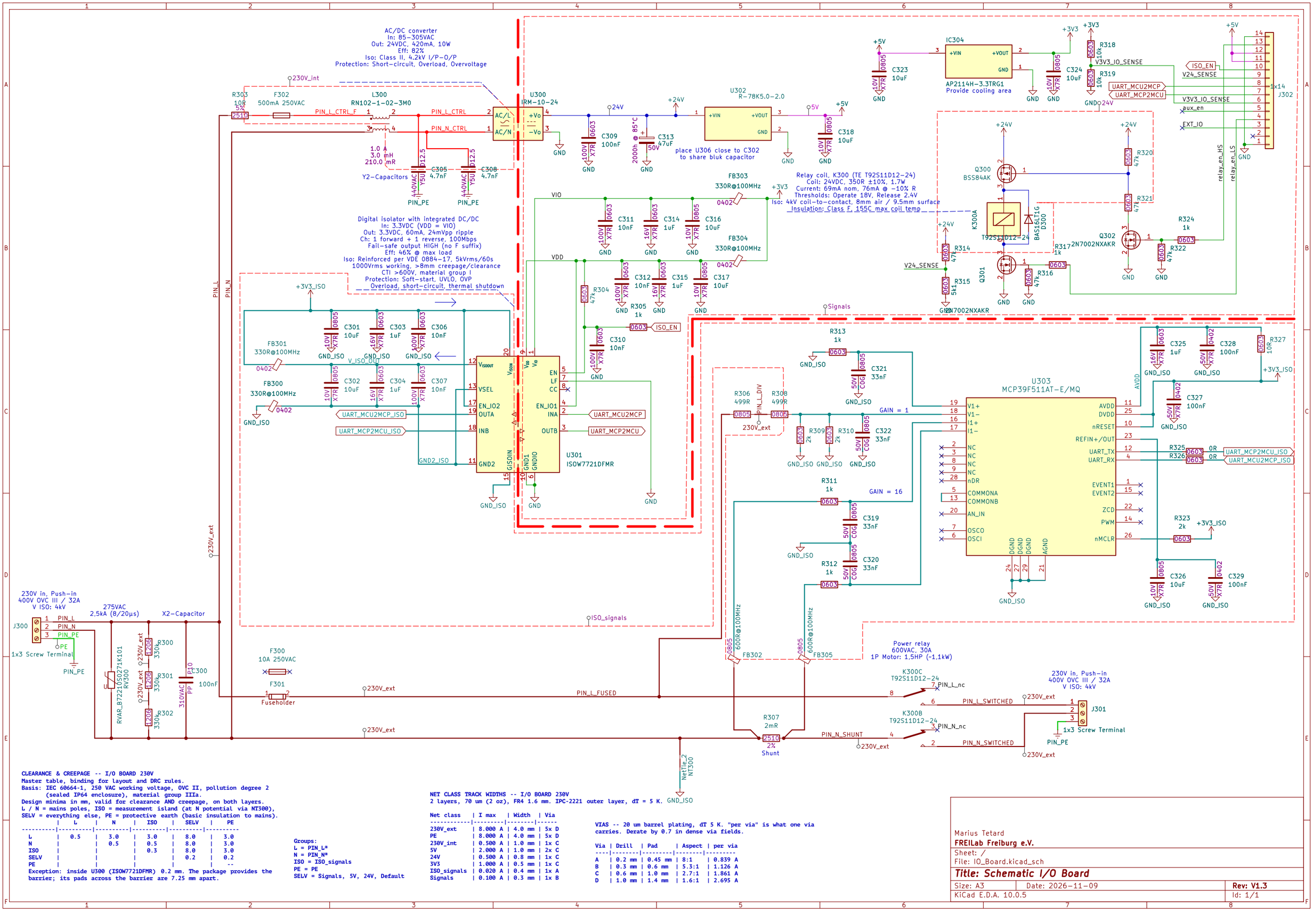


CLEARANCE & CREEPAGE -- I/O BOARD 230V
Master table, binding for layout and DRC rules.
Basis: IEC 60664-1, 250 VAC working voltage, OVC II, pollution degree 2
(sealed IP64 enclosure), material group IIIa.
Design minima in mm, valid for clearance AND creepage, on both layers.
L / N = mains poles, ISO = measurement island (at N potential via NT300),
SELV = everything else, PE = protective earth (basic insulation to mains).

	L	N	ISO	SELV	PE
L	0.5	3.0	3.0	8.0	3.0
N		0.5	0.5	8.0	3.0
ISO			0.3	8.0	3.0
SELV				0.2	0.2
PE					--

Exception: inside U300 (ISOW7721DFMR) 0.2 mm. The package provides the barrier; its pads across the barrier are 7.25 mm apart.

Groups:
L = PIN_L*
N = PIN_N*
ISO = ISO_signals
PE = PE
SELV = Signals, 5V, 24V, Default

NET CLASS TRACK WIDTHS -- I/O BOARD 230V
2 layers, 70 um (2 oz), FR4 1.6 mm. IPC-2221 outer layer, dT = 5 K.

Net class	I max	Width	Via
230V_ext	8.000 A	4.0 mm	5x D
PE	8.000 A	4.0 mm	5x D
230V_int	0.500 A	1.0 mm	1x C
5V	2.000 A	1.0 mm	2x C
24V	0.500 A	0.8 mm	1x C
3V3	1.000 A	0.5 mm	1x C
ISO_signals	0.020 A	0.4 mm	1x A
Signals	0.100 A	0.3 mm	1x B

VIAS -- 20 um barrel plating, dT 5 K. "per via" is what one via carries. Derate by 0.7 in dense via fields.

Via	Drill	Pad	Aspect	per via
A	0.2 mm	0.45 mm	8:1	0.839 A
B	0.3 mm	0.6 mm	5.3:1	1.126 A
C	0.6 mm	1.0 mm	2.7:1	1.861 A
D	1.0 mm	1.4 mm	1.6:1	2.695 A

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Title: Schematic I/O Board

Size: A3	Date: 2026–11–09	Rev: V1.3
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